Nielsen Bookscan Report

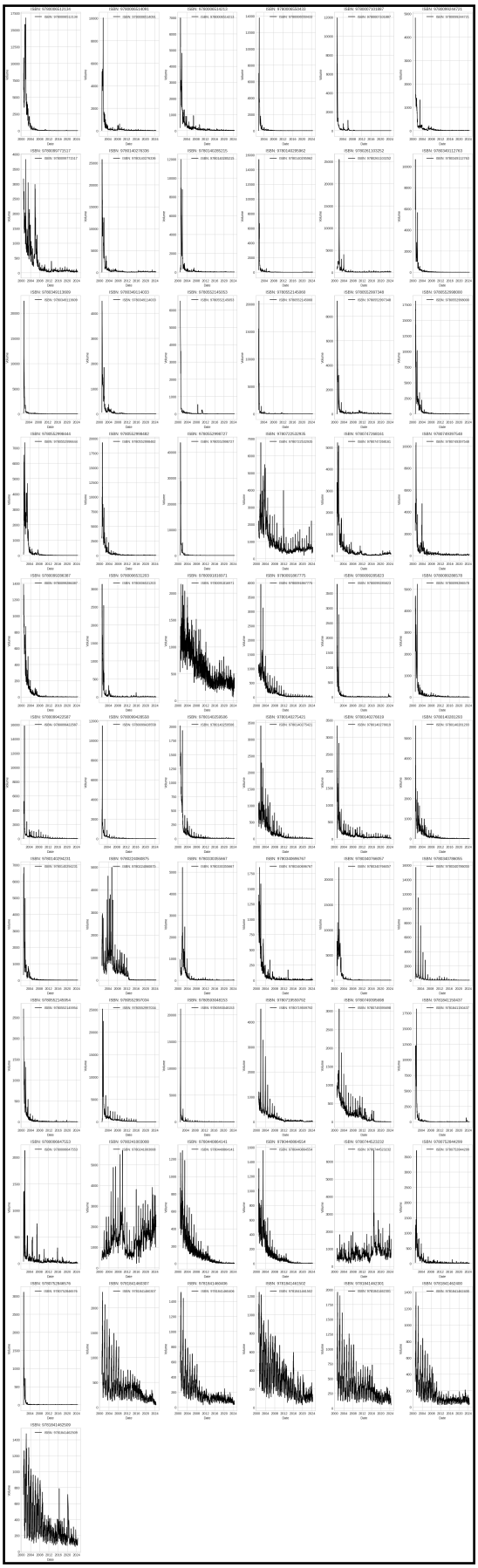
9.1: Using time-series analysis for sales and demand forecasting

Sarah Stapleton
CAM003
7 September 2025

Abstract

A time-series workflow was implemented to analyse weekly book sales and it commences with ISBNs to identify sales data after 2012-01-01 and visualises the full set (Figure 1). The analyses generate short-term forecasts for two titles—*The Alchemist* and *The Very Hungry Caterpillar*. Classical diagnostics (STL decomposition; ACF/PACF; ADF) motivate further modelling choices: for strong yearly periodicity of *The Alchemist* where the series is stationary; and for, *Caterpillar* differencing is necessitated. Modelling proceeds with (i) Auto-ARIMA/SARIMAX, (ii) an ML reduction using XGBoost for monthly aggregation, and (iii) a SARIMA+LSTM hybrid (sequential and parallel). Monthly forecasts are compared between XGBoost and Auto-ARIMA with an aggregated weekly SARIMA baseline.

Figure 1: ISBN of 61 Book Sales (2012-01-01 — 2024-07-01)

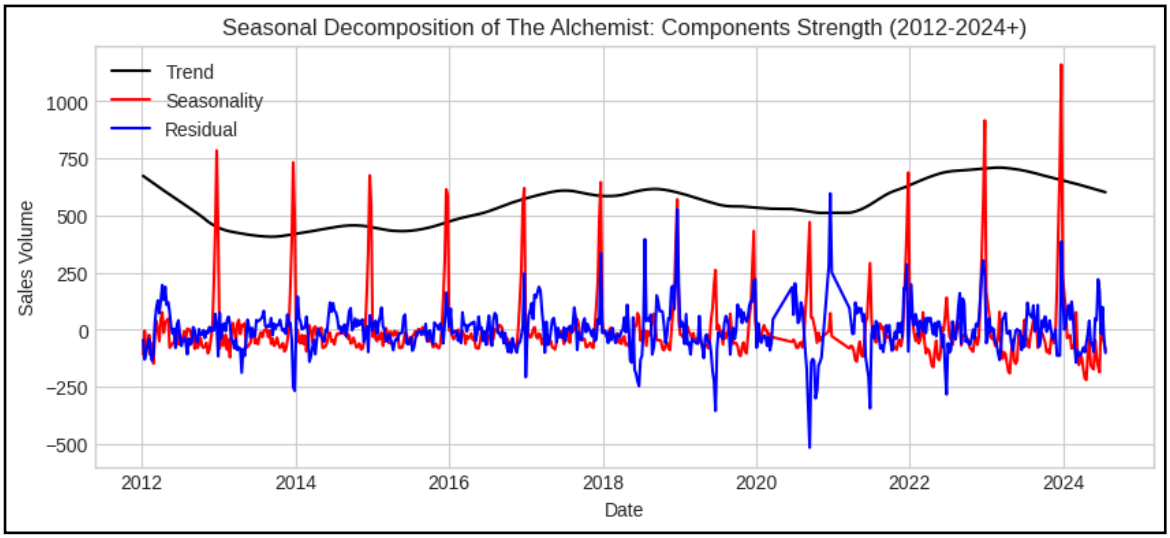


1.0 Introduction

The Nielsen Bookscan sales data requires a granular approach to preserve natural cadence and resolve false-positives from outliers and skew. The short-term horizon demand forecast for inventory and publishing decision-making required data coercion to resolve non-Gaussian variance and remove ambiguity across time-periods. This project addresses uncertainty and classical models fit on the data and reports MAE/MAPE at each stage to communicate results. The data uses 61 ISBNs to plot sales that characterise heterogeneity (Figure 1), and two books sustain signals with interpretable structures; The Alchemist and The Very Hungry Caterpillar. Prior to transformation, The Alchemist exhibited skew 2.895 and became nearly Gaussian (0.108), while Caterpillar was observed at 0.691 and -1.153, a left-skew; Caterpillar's sales continue to strengthen.

2.0 Seasonal Affects in Time-series Analysis

A Seasonal-Trend-Linearity (STL) decomposition (Figures 2–3) is a model-agnostic method that distinguishes between components influencing sales with stationarity and seasonality. ADF tests analysed the unit root and showed *Alchemist* exhibits evidence for stationarity (p≈0.00, ADF=-9.793) without differencing, trend = 0.48 and stable seasonality on an annual basis; residuals have modest effect. Conversely, *Caterpillar* is non-stationary (p≈0.2061, ADF=-2.2), has a strong trend (0.791) and seasonal strength (0.664); residual variation suggests intermittent boosts in sales, potentially attributable to gift seasons. Price remains continuous for both books.



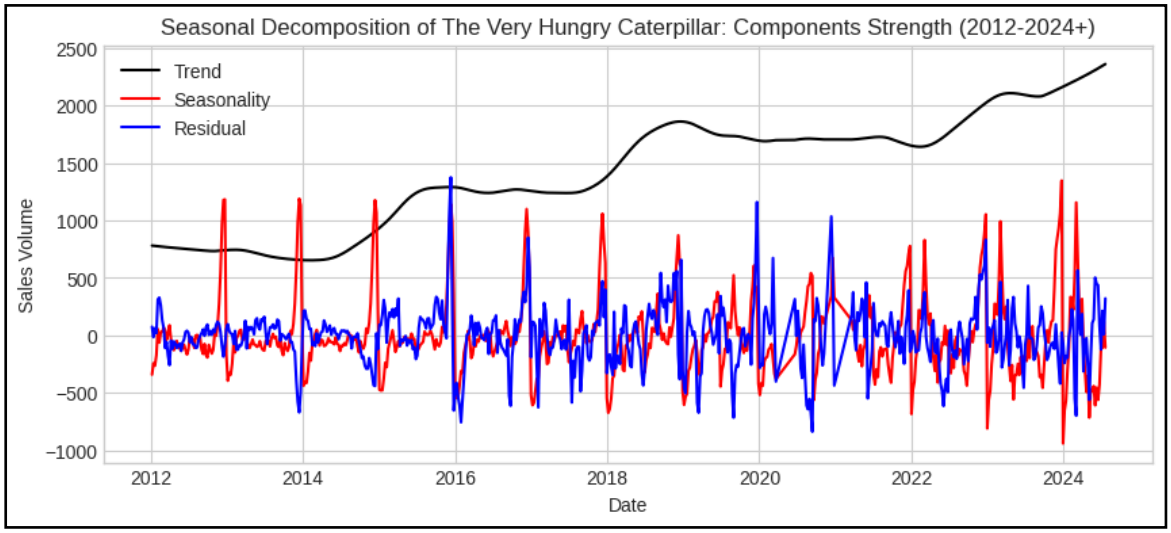


Figure 2–3: STL Decomposition

Canova-Hansen was implemented via auto-ARIMA to confirm sesonality/stationarity and confirm a differencing outcome of d=0 and d=1 respectively.

3.0 Modelling Methods and Interpretation

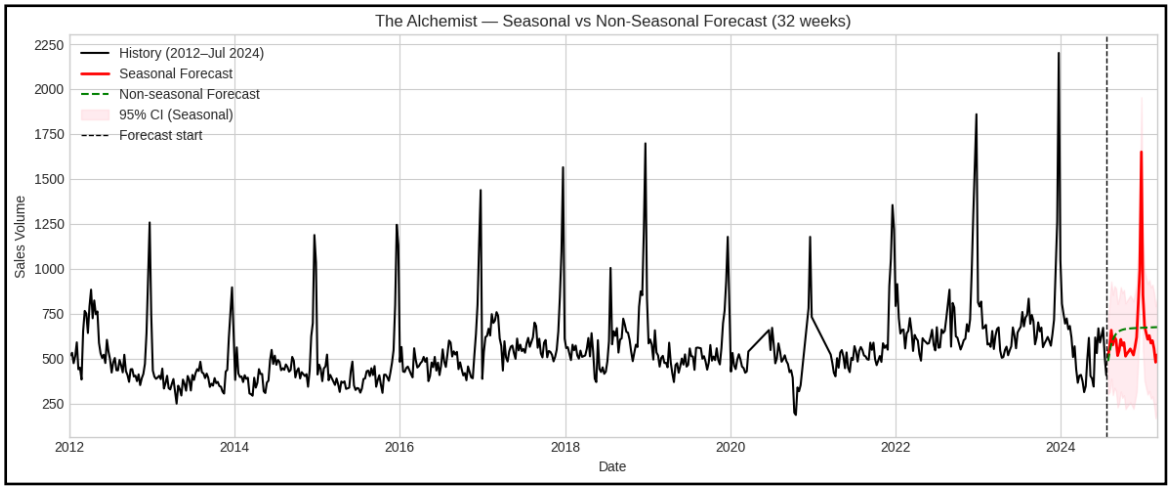
```
Alchemist:
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                   y   No. Observations:                  628
Model:             SARIMAX(1, 0, 2)x(1, 0, [], 52)   Log Likelihood               -3842.827
Date:                              Sun, 14 Sep 2025   AIC                           7697.654
Time:                                      14:14:24   BIC                           7724.309
Sample:                                           0   HQIC                          7708.009
                                              - 628
Covariance Type:                                opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
intercept     41.9341     10.655      3.935      0.000      21.050      62.818
ar.L1          0.7767      0.051     15.311      0.000       0.677       0.876
ma.L1         -0.1127      0.057     -1.960      0.050      -0.225   -2.83e-05
ma.L2         -0.0867      0.059     -1.472      0.141      -0.202       0.029
ar.S.L52       0.6649      0.022     30.117      0.000       0.622       0.708
sigma2      1.208e+04    434.542     27.789      0.000    1.12e+04    1.29e+04
===================================================================================
Ljung-Box (L1) (Q):                   0.06   Jarque-Bera (JB):              1949.60
Prob(Q):                              0.81   Prob(JB):                         0.00
Heteroskedasticity (H):               2.64   Skew:                             1.23
Prob(H) (two-sided):                  0.00   Kurtosis:                        11.27
===================================================================================
```

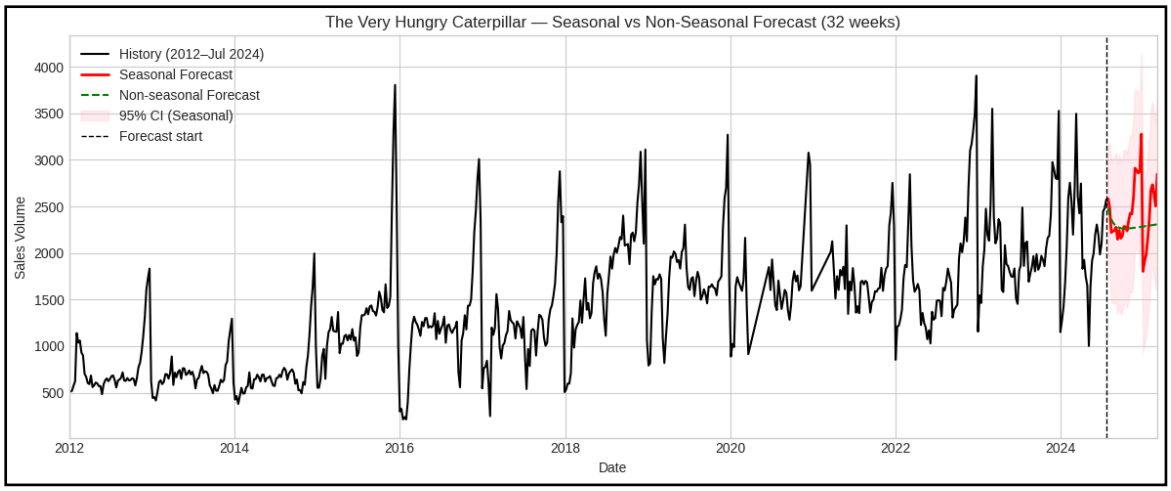
The SARIMAX models described below are produced by Auto-Arima.

```
Caterpillar:
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                   y   No. Observations:                  628
Model:             SARIMAX(1, 1, 1)x(1, 0, [1, 2], 52)   Log Likelihood               -4433.438
Date:                                Fri, 12 Sep 2025   AIC                           8878.877
Time:                                        12:22:09   BIC                           8905.522
Sample:                                             0   HQIC                          8889.229
                                                  - 628
Covariance Type:                                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.6580      0.029     22.912      0.000       0.602       0.714
ma.L1         -0.9540      0.012    -77.298      0.000      -0.978      -0.930
ar.S.L52       0.4918      0.073      6.761      0.000       0.349       0.634
ma.S.L52      -0.0437      0.084     -0.520      0.603      -0.209       0.121
ma.S.L104      0.1873      0.053      3.563      0.000       0.084       0.290
sigma2      7.839e+04   2594.428     30.215      0.000    7.33e+04    8.35e+04
===================================================================================
Ljung-Box (L1) (Q):                   0.24   Jarque-Bera (JB):               835.38
Prob(Q):                              0.62   Prob(JB):                         0.00
Heteroskedasticity (H):               3.08   Skew:                            -0.19
Prob(H) (two-sided):                  0.00   Kurtosis:                         8.64
===================================================================================
```

3.1 SARIMAX

Figure 5-6: Seasonal vs Non-Seasonal Products of SARIMAX with Forecasts (32 weeks)

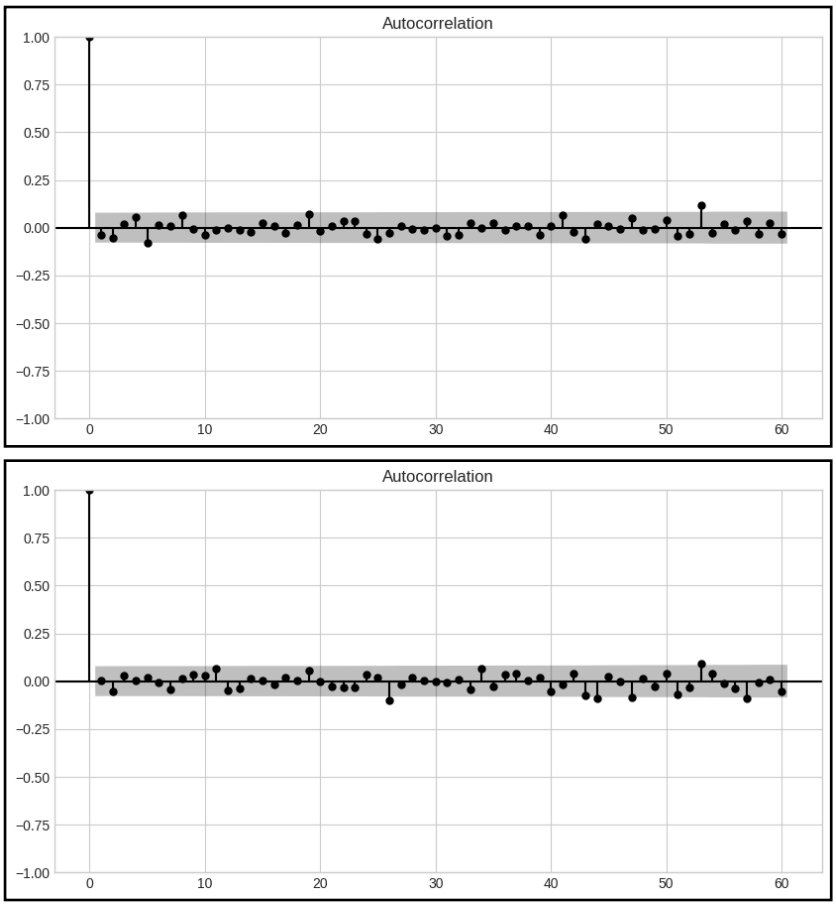




A comparative analysis of seasonal SARIMA vs non-seasonal ARIMA on The Alchemist with structural calibration of **SARIMAX(1,0,2)x(1,0,0,52)** deduces key metrics that anticipate stable auto-regressive seasonality; intercept ≈ 41.9 grounds a baseline, Auto-Regressive (L1)=0.777 at lag=52 current sales dependent on prior weeks indicated annually, weak MA(L1)=−0.1127, immense seasonal continuity AR(52)=0.665; $\sigma^2 \approx 1.2k$ residuals. Moreover, additional modelling methods are implemented as a result of distortions and residual heteroskedasticity in spite of the general reliability of SARIMAX.

An identical test on Caterpillar **SARIMAX(1,1,1)x(1,0,[1,2],52)** exposed a strong 66% of sales depend on the prior week, especially in Winter, with a repeated error every two years (MA(L2=0.187) and estimated variance of residuals $\sigma^2 \approx 0.784k$. The high **Kurtosis = 8.64** accompanies a fat-tailed-residual distribution of uncorrelated residuals in patterns.

Figure 4-5: Autocrrelation of Residual Values: The Alchemist (top) and The Very Hungry Caterpillar (bottom)

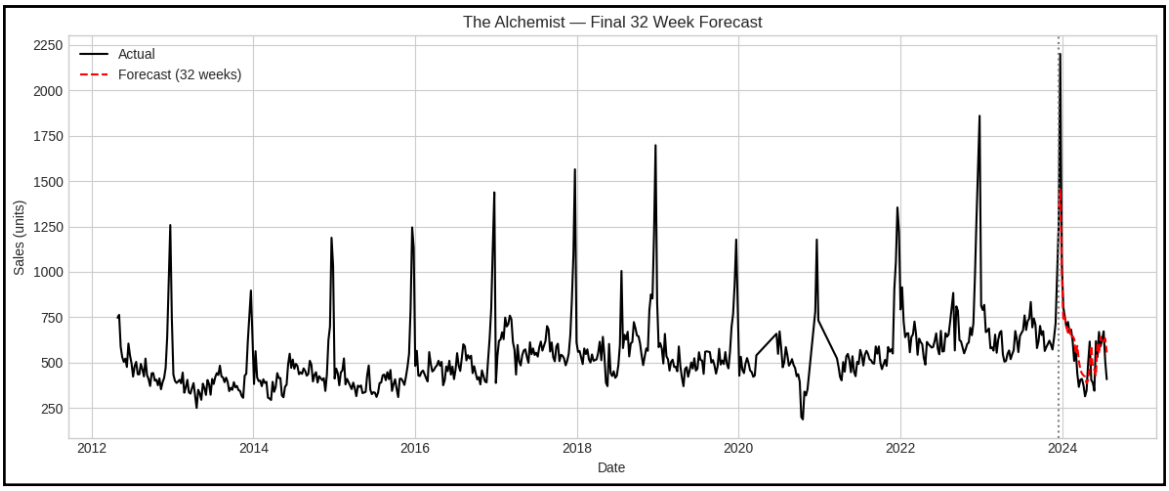


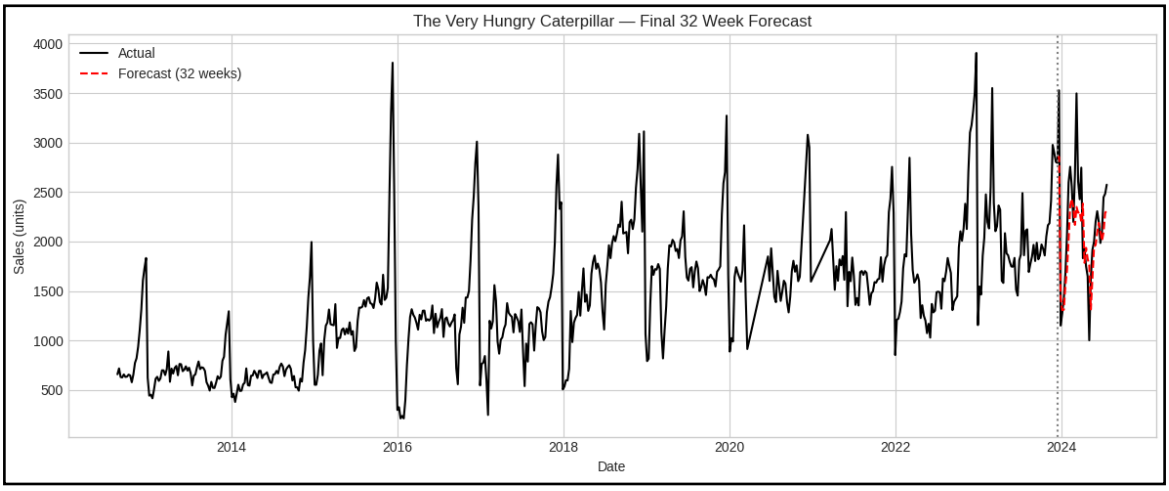
A residual auto-correlative plot (figs.

4-5) evidences some model error with some spikes above shaded confidence boundaries.

3.2 XGBoost

Figure 7-8: XGBoost



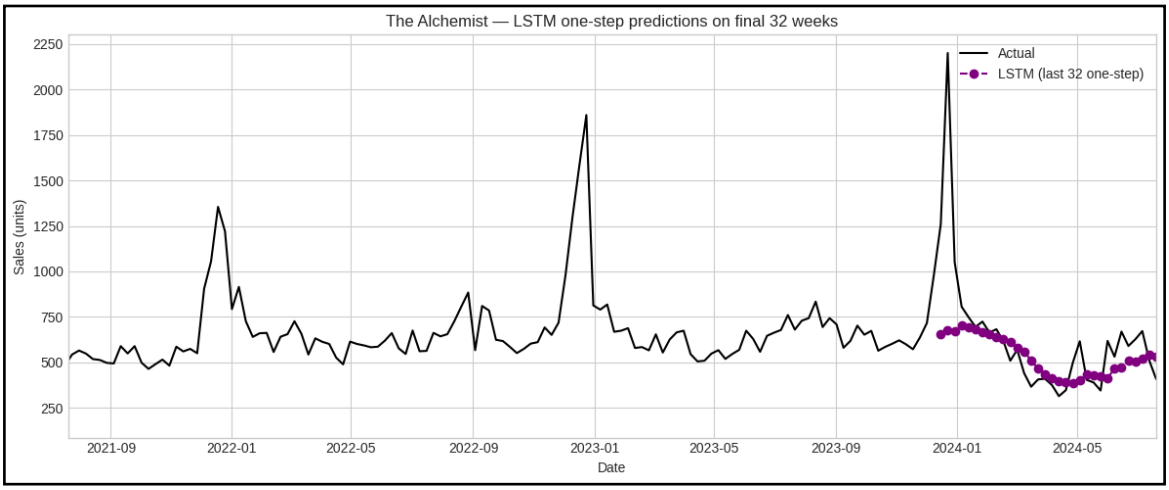


The XGB's model received pre-processed, lag-based, feature-engineered data and seasonality variables, split by temporal periods into series to preserve datetime indices. The XGBoost model's parameter tuning and grid search identified a superior model for a 32-week forecast. It confirms smoothly inclining spikes that indicate *no seasonal features in recursive week-to-week plots* (Alchemist: MAE 101.15, MAPE 17.05%; Caterpillar: MAE 297.29, MAPE 14.2%).

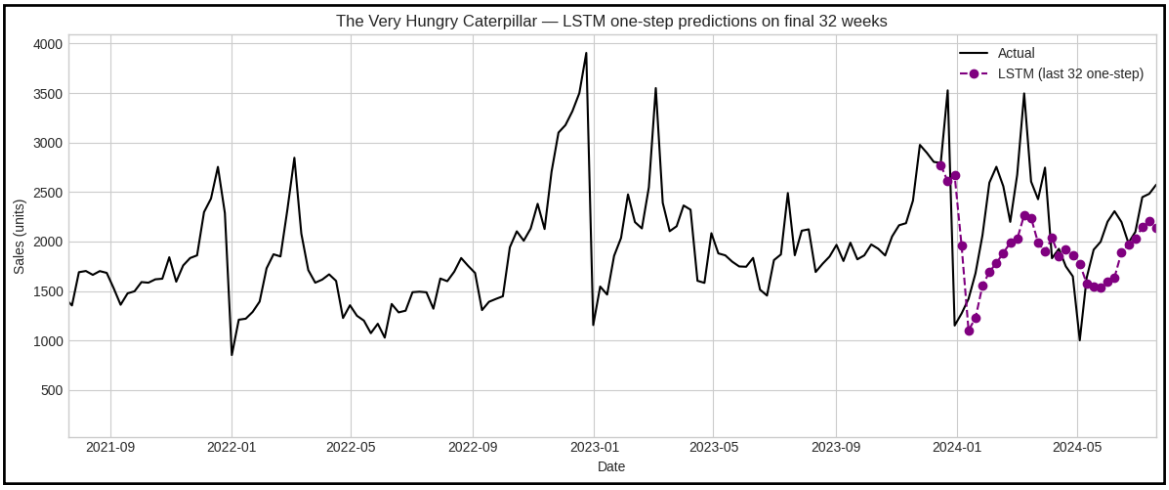
3.3 LSTM

Experimentation implemented Grid-Search-CV to yield optimum scaling with favourable results from training with a 'looking-back at history' window=16 because it has the lowest validation loss **(*Alchemist:*** MAE 155, MAPE 19.36%; ***Caterpillar:*** MAE 491.3, MAPE 24.71%). The forecasts deviate from actuals by an average 155 books sold per week for The Alchemist and 491 books for the Caterpillar derived from MAE on the same scale as the data; MAPE reports the results as % that normalize residuals, such that *a relative error spans fewer units* for Caterpillar. Overall, the plots reveal recent history has the greatest sales implications.

Figs. 9-10: LSTM Forecasts



The Alchemist: lookback=16, val_loss=0.7784, MAE: 153.01, MAPE: 19.36%

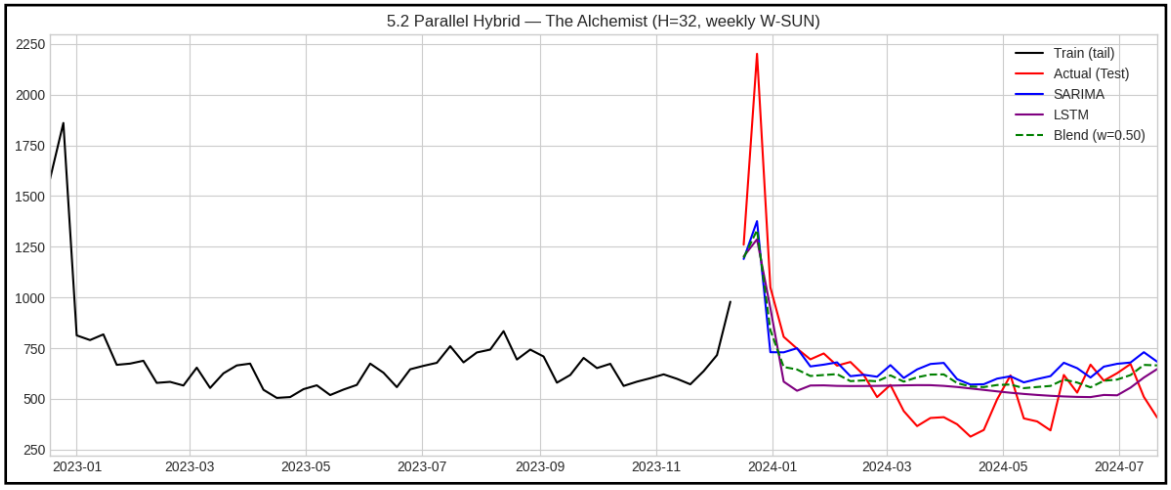


The Very Hungry Caterpillar: lookback=24, val_loss=0.6633, MAE: 491.30, MAPE: 24.71%

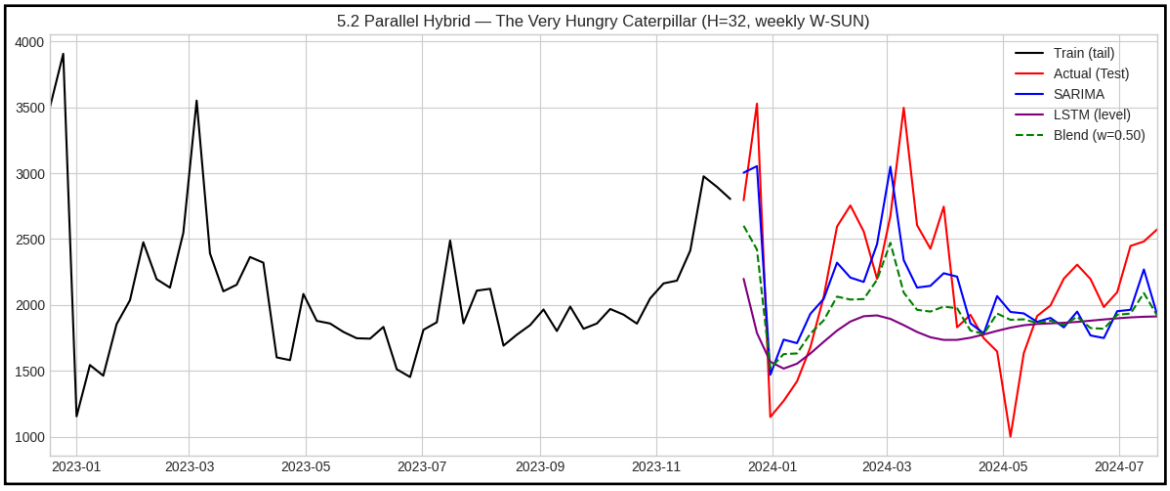
3.4 Hybrid Modelling (SARIMA + LSTM)

The hybrid model overlays SARIMA (statistical inductive bias) with LSTM (nonlinear residual structure). The plot offers a forecast-start delimiter and reports MAE/MAPE for SARIMA, LSTM, and a blend of them. These metrics and plots affirm that low hybridisation can effectively smooth the systematic bias of SARIMA while retaining seasonality. The w* was chosen through a validation of MAE over **w∈{0.0,...,1.0}** from a grid of 11 possibilities; this step is called fine-tuning.

Figure 11-12: Parallel Hybrid SARIMA+LSTM (w=0.5) with Independent SARIMA and LSTM



The Alchemist — Parallel Hybrid (w=0.50)
SARIMA : MAE=155.34, MAPE=30.15%
LSTM : MAE=151.5, MAPE=26.58%
Hybrid Blended Model (w=0.5): MAE=146.36, MAPE=27.2%

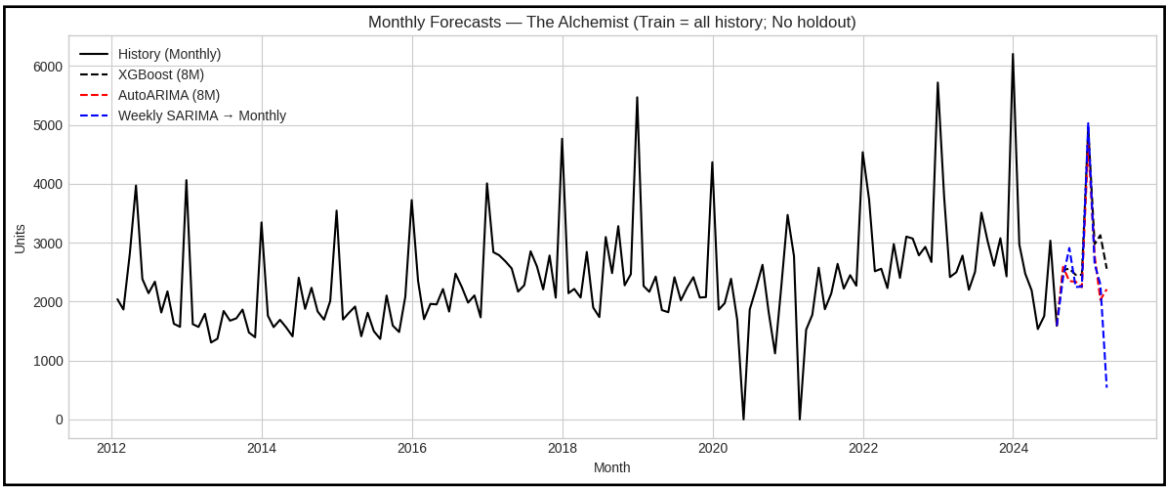


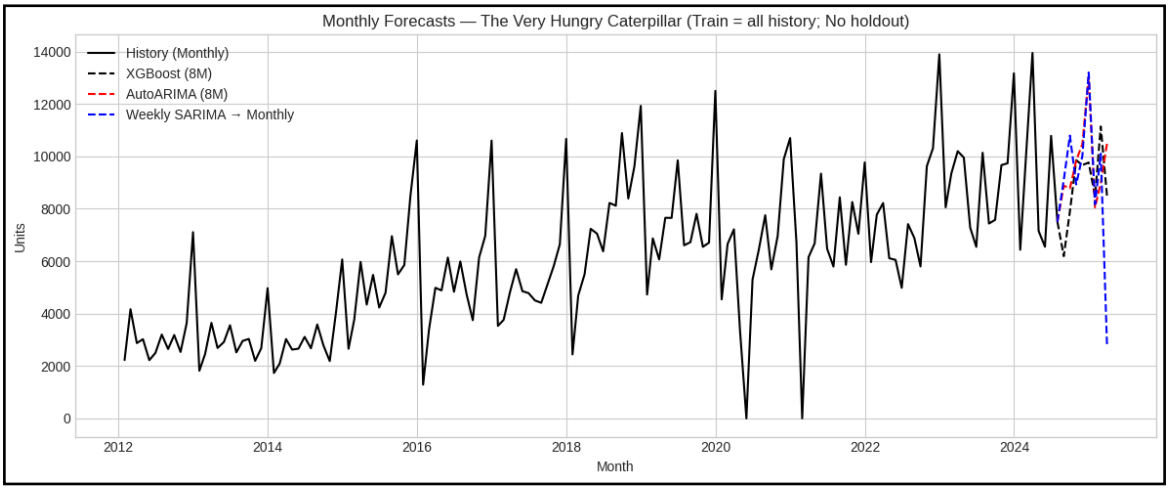
The Very Hungry Caterpillar — Parallel Hybrid (w=0.50)
SARIMA : MAE=357.88, MAPE=17.8%
LSTM : MAE=495.86, MAPE=21.6%
Blend (w=0.5): MAE=395.98, MAPE=18.41%

3.5 Monthly Aggregation Comparison

In the final section, monthly aggregations were trained on SARIMA residuals; XGBoost works on a lagged supervised table, Auto-ARIMA on the monthly series.

Figure 9-10: Paralle Hybrid Blend of SARIMA + LSTM on Monthly Sales Data





This analysis has provided strong justification for a multi-modal approach to time-series forecasting. Firstly, the SARIMAX/Auto-ARIMA's strong baseline, interpretability, and ability to leverage seasonal data structures is inferred by ACF/PACF/STL plots with low variance. Secondly, LSTM indicates flexible nonlinear modelling that captured contextual differences, or residual patterns escapable from SARIMAX. Thirdly, XGBoost uses monthly lags for rapid, aggregated baseline horizons, offering robust performance with limited data. A compliment of different approaches, this robust model evades biases/variance distortions.

	Alchemist					Caterpillar				
	MAE	MAPE%	MeanResid	StdResid	RMSE	MAE	MAPE%	MeanResid	StdResid	RMSE
Model										
XGBoost (M)	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
AutoARIMA (M)	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
Weekly SARIMA→Monthly	798.23	-15.91	-29.55	1254.77	1174.10	1835.84	3.24	788.47	2346.51	2332.28

4.0 Discussion

'The Alchemist' has minimal structural change over time, a stable book barely affected by seasonal sales, while Caterpillar has an upward trend, and seasonal gravity (trend strength = 0.79). The Ljung-Box tests describes temporal, structural representations and that suggests a Confidence Interval misaligned with actuals. Tests implemented are positive but protect against optimistic bias, observed in the use of Keras tuning and Grid-Search-CV optimisation as shown by MAE/MAPE. Where interval periods dominate decision-making, tests on residuals using GARCH could support reliable, probabilistic inventory forecasts. Furthermore, the sustained trend, strong seasonality and robust margins suggests that re-publication would support consumer demand. Caterpillar's positive trend and seasonal spikes are indicative of gift purchasing with potential for re-published volumes.

5.0 Conclusion

The analysis follows a defensible diagnostic. Seasonal SARIMAX offers a reliable weekly baseline; the hybrid-model suggests the potential for incremental accuracy; and the monthly XGBoost vs Auto-ARIMA comparison provides a pragmatic operational view for monthly planning. The improved plotting strategy and explicit dates used support traceability. This approach justifies a recommendation for book-specific analysis selecting lowest MAE/MAPE scores and validated by an expanding-window CV.

This report pre-processes genuine Nielsen data, diagnoses statistical properties, decomposes signal, implements and tunes multiple forecasting models, with backtests and diagnostics for validation purposes, to enable inventory/reprint decisions for SME publishers—prioritising titles with strong seasonal signatures and positive trends for higher-margin, lower-risk investment.